reference frame. When gravity is present: this same principle, but with local inertial reference frames playing the role of the inertial reference frames.

pulsar. A magnetized, spinning neutron star that emits a beam of radiation (radio waves and sometimes also light and X-rays). As the star spins, its beam sweeps around like the beam of a turning spotlight; each time the beam sweeps past Earth, astronomers receive a pulse of radiation.

pulsation. The vibration or oscillation of an object, for example, a black hole or a star or a bell.

quantum field. A field that is governed by the laws of quantum mechanics. All fields, when measured with sufficient accuracy, turn out to be quantum fields; but when measured with modest accuracy, they may behave classically (that is, they do not exhibit wave/particle duality or vacuum fluctuations).

quantum fields in curved spacetime, the laws of. A partial marriage of general relativity (curved spacetime) with the laws of quantum fields, in which gravitational waves and nongravitational fields are regarded as quantum mechanical, while the curved spacetime in which they reside is regarded as classical.

quantum foam. A probabilistic foamlike structure of space that probably makes up the cores of singularities, and that probably occurs in ordinary space on scales of the Planck—Wheeler length and less.

quantum gravity. The laws of physics that are obtained by merging ("marrying") general relativity with quantum mechanics.

quantum mechanics. The laws of physics that govern the realm of the small (atoms, molecules, electron, protons), and that also underlie the realm of the large, but rarely show themselves there. Among the phenomena that quantum mechanics predicts are the *uncertainty principle, wave/particle duality,* and *vacuum fluctuations.*

quantum nondemolition. A method of measurement that circumvents the standard quantum limit.

quantum theory. The same as *quantum mechanics.*

quasar. A compact, highly luminous object in the distant Universe, believed to be powered by a gigantic black hole.

radiation. Any form of high-speed particles or waves.